STEP-INSPECTOR: DECODING-FREE REASONING ERROR LOCALIZATION VIA STEP-LEVEL REPRESENTATION PROBING IN SMALL LANGUAGE MODELS

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ABSTRACT

Large Language Models (LLMs) are widely deployed as reference-free evaluators (*LLM-as-a-Judge*), yet prompt-based critique generation suffers from high autoregressive decoding latency, format non-compliance in smaller models, and sensitivity to position and verbosity biases. Recently, *INSPECTOR* (Li et al., 2026) validated the **Semantic Capacity Asymmetry Hypothesis**, showing that small language models (SLMs) encode rich evaluative signals in their internal representations despite weak generative ability. However, existing *Representation-as-a-Judge* approaches operate only at coarse, whole-response granularity, averaging out localized arithmetic or logical failures in multi-step reasoning. In this work, we introduce **Step-INSPECTOR**, an empirical investigation into whether frozen SLM representations can achieve *decoding-free step-level error localization*. Using step-annotated solutions from **MR-GSM8K** across two frozen backbones—`Qwen2.5-0.5B-Instruct` (24 layers) and `Qwen3-0.6B` (28 layers)—we extract token-aligned step representations via character-offset mapping and train lightweight probes using a **Causal Masked Step Binary Cross-Entropy Loss** ($\mathcal{L}_{\text{causal}}$) that masks post-error steps ($m > k^*$) to prevent false-premise contamination. In pooling ablations, whole-response global pooling is mathematically capped at 10.0% First-Error Accuracy (4.4%–5.1% mean across layers), whereas step-span mean pooling reaches 38.0%–40.0% peak First-Error Accuracy. Under a strict 5-fold out-of-fold (OOF) evaluation protocol, Step-INSPECTOR achieves test Step AUROCs of **0.880** $[0.84, 0.92]$ on `Qwen2.5-0.5B` (Layer 19) and **0.857** $[0.80, 0.90]$ on `Qwen3-0.6B` (Layer 16), substantially outperforming content-free position-only (0.506) and length-only (0.708) baselines. At the same time, solution-clustered bootstrap audits reveal the statistical limits of 100-example evaluation: while top-step localization (38.0% and 36.0%) nearly triples random guessing (13.5%), its paired confidence interval against length-only (24.0%) includes zero, and joint accuracy at the balanced operating point (55.0% and 52.0%) does not statistically exceed the 50.0% trivial baseline on a balanced split. Code and evaluation scripts are available at `https://github.com/thejesshwar/step_inspector`.

[*]Member 1: Spatial & Pooling Dynamics Track.
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1 INTRODUCTION

Evaluating multi-step mathematical and logical reasoning is a central bottleneck in scaling reliable Large Language Models (LLMs) (Li et al., 2024; Lightman et al., 2023). While the *LLM-as-a-Judge* paradigm (Zheng et al., 2023) replaces costly human annotation with prompted model critiques, generative evaluation exhibits three fundamental limitations. First, autoregressive decoding of step-by-step critiques incurs substantial latency and compute overhead, requiring hundreds of sequential token forward passes per candidate solution. Second, prompt-based judges are vulnerable to presentation shortcuts, including position bias and verbosity/length bias (Dubois et al., 2024; Ye et al., 2024). Third, when attempting to reduce compute by deploying Small Language Models (SLMs, < 1B parameters) as generative critics—even with parameter-efficient soft-prompt tuning (Lester et al., 2021) or Critique Fine-Tuning (Wang et al., 2025)—models frequently fail strict output formatting constraints (producing unparseable `VERDICT` or `FIRST_ERROR` headers) or hallucinate invalid justifications.

To overcome the bottlenecks of generative evaluation, Li et al. (2026) recently proposed the **Semantic Capacity Asymmetry Hypothesis**: *the semantic capacity required for accurate evaluation is significantly lower than that required for surface text generation, and evaluative signals are already linearly or shallowly accessible in the intermediate hidden states of small language models.* Instantiating this insight as *Representation-as-a-Judge* (**INSPECTOR**), Li et al. (2026) demonstrated that probing frozen SLM representations substantially outperforms prompted SLM critiques across GSM8K, MATH, and GPQA.

The Problem: Granularity Collapse in Whole-Response Probing. Despite its efficiency, base INSPECTOR extracts a single pooled vector over the entire candidate response to predict coarse, response-level quality scores. In multi-step mathematical reasoning (Cobbe et al., 2021; Zeng et al., 2024), a candidate solution $Y_i = (s_1, s_2, \ldots, s_{M_i})$ typically contains several valid derivation steps alongside a single arithmetic or conceptual slip at step k_i^*. Global pooling across hundreds of tokens dilutes localized error signals, rendering whole-response probes incapable of identifying *where* the reasoning chain breaks down. Furthermore, supervising step-level probes introduces a subtle causal dilemma: once a deductive error occurs at step k_i^*, all subsequent steps ($m > k_i^*$) operate on corrupted premises. Naively labeling post-error steps as valid ($y_m = 0$) or faulty ($y_m = 1$) injects contradictory supervision gradients into the probe.

Our Contributions. We propose **Step-INSPECTOR**, extending the *Representation-as-a-Judge* framework from whole-response scoring to decoding-free, step-level error localization on the MR-GSM8K benchmark (Zeng et al., 2024). Requiring only a single non-autoregressive prefill pass (0 generated tokens and 100% format compliance by construction), Step-INSPECTOR is evaluated across two frozen open-weight SLMs—`Qwen2.5-0.5B-Instruct` (Yang et al., 2024) and `Qwen3-0.6B` (Yang et al., 2025)—through a four-track empirical investigation:

1. **Spatial & Pooling Dynamics (Track 1, Member 1):** We build a deterministic character-to-token span slicer (verified on $650/650$ MR-GSM8K solutions) and compare step-wise *Mean Pooling*, *Last-Token Pooling*, and base INSPECTOR *Global Whole-Response Pooling*. We show mathematically and empirically that global pooling collapses on error localization (4.4%–5.1% mean First-Error Accuracy across layers, upper-bounded by the 10.0% Step-1 error rate), whereas step-span pooling recovers up to 38.0%–40.0% peak First-Error Accuracy.
2. **Probe Capacity & Separability Dynamics (Track 2, Member 2):** We conduct a multi-seed capacity ablation comparing a parameter-minimal `LinearProbe` ($d + 1$ parameters) against a bottleneck `MLPProbe` with LayerNorm and GELU across confusion-matrix diagnostics. We show that non-linear capacity improves error recall and localization on `Qwen2.5-0.5B` ($+8.67$ pp First-Error Accuracy) but offers no localization gain on `Qwen3-0.6B` (-0.67 pp), indicating stronger linear separability in `Qwen3-0.6B`.
3. **Layer-Depth Geometry & Causal Supervision Masking (Track 3, Member 3):** We probe all 24 and 28 layers of both backbones, compare single-layer probes against a parametric `CrossLayerAttentionProbe` (CLAP), and formulate the **Causal Masked Step BCE Loss** ($\mathcal{L}_{\text{causal}}$). We show that evaluative signal plateaus in mid-to-late layers (~ 0.57–0.83 relative depth) and that causal masking boosts First-Error Accuracy by $+14.0$ to $+20.0$ pp

over naive unmasked (Target $= 0$) supervision while maximizing Joint Accuracy (59.0%–60.0%).

4. **Diagnostic Benchmarking & Bias Auditing (Track 4, Member 4):** Using a leakage-free 5-fold out-of-fold (OOF) cross-validation protocol with solution-clustered bootstrap confidence intervals, we benchmark Step-INSPECTOR against content-free *Position-Only* and *Length-Only* heuristics, audit score-length correlations across length tertiles, and execute a paired adversarial padding stress test.

2 Related Work

LLM-as-a-Judge and Process Verification. Using LLMs as reference-free evaluators has become standard for open-ended generation and reasoning (Zheng et al., 2023; Liu et al., 2023; Li et al., 2024). In mathematical reasoning, Lightman et al. (2023) showed that step-level process supervision (PRMs) outperforms outcome supervision, while **MR-GSM8K** (Zeng et al., 2024) established a meta-reasoning benchmark requiring evaluators to predict both solution-level verdicts and the exact index k^* of the first reasoning error. Recently, Critique Fine-Tuning (CFT) (Wang et al., 2025) trained models to generate natural language critiques. However, generative judges remain computationally expensive and suffer from position bias, verbosity bias, and format fragility in sub-billion-parameter regimes (Ye et al., 2024; Dubois et al., 2024).

Probing Internal Representations and INSPECTOR. Linear and non-linear probing of intermediate Transformer representations has revealed that hidden states encode syntactic structure (Rogers et al., 2020), world states (Li et al., 2023), and latent truth value even when surface generations are false (Burns et al., 2023; Kadavath et al., 2022). Kantamneni et al. (2025) further demonstrated that simple mean-pooled linear probes remain highly competitive against complex sparse autoencoders. Bridging probing with evaluation, **INSPECTOR** (Li et al., 2026) introduced *Representation-as-a-Judge*, showing that probing frozen SLMs (`Qwen3-0.6B`, `Qwen3-1.7B`, `Llama-3.2-1B`) approximates DeepSeek-V3 aspect scores at a fraction of the compute. Step-INSPECTOR directly extends Li et al. (2026) from whole-response aspect classification to fine-grained, step-by-step mathematical error localization.

Distinction from Self-Verification Probing (*ReProbe*). Recent work such as *ReProbe* (Ni et al., 2026) and confidence-calibration probes (Kadavath et al., 2022; Ren et al., 2023) probes a generator's *own* hidden states during autoregressive decoding to guide test-time scaling (e.g., Best-of-N and Beam Search). Specifically, Ni et al. (2026) train a $<$ 10M-parameter Transformer encoder over internal states to estimate step credibility during generation, and in their token-level linear baseline label all tokens of an incorrect step without addressing post-error premise contamination across complete external trajectories. Step-INSPECTOR differs along three key dimensions: (1) **External Candidate Verification:** rather than introspecting on self-generated rollouts during decoding, Step-INSPECTOR acts as an external judge that verifies arbitrary candidate solutions in a single prefill forward pass; (2) **Parameter-Minimal Step-Span Pooling:** rather than a multi-megabyte Transformer probe, we apply parameter-free token-span mean pooling followed by a 1K–260K parameter probe head; and (3) **Causal Post-Error Truncation ($\mathcal{L}_{\text{causal}}$):** we explicitly mask post-error steps ($m > k^*$) to isolate the earliest deductive failure k^* and audit against position and length shortcuts.

3 Methodology

3.1 Problem Formulation, Step-Span Slicing, and Spatial Pooling

Let Q_i denote an arithmetic word problem and $Y_i = (s_1, s_2, \ldots, s_{M_i})$ denote a candidate solution partitioned into $M_i \leq M_{\max}$ sequential reasoning steps (with $M_{\max} = 17$ across the dataset). Each solution carries a binary verdict $V_i \in \{\text{CORRECT}, \text{INCORRECT}\}$ and, if incorrect, a ground-truth first-error step index $k_i^* \in \{1, \ldots, M_i\}$ indicating the earliest substantive mathematical or logical fallacy (with $k_i^* = 0$ if $V_i = \text{CORRECT}$).

Deterministic Step-Span Token Mapping. We format (Q_i, Y_i) using the backbone's chat template with system prompt "`You solve arithmetic word problems accurately.`" (setting `enable_thinking=False` on `Qwen3-0.6B` to prevent empty `<think>` blocks from

shifting character offsets). To map character-level step boundaries in Y_i to exact token indices in the tokenized prompt P_i, we apply the regular expression:

```
r"(Step\s+\d+[:\.\-].*?)(?=(?:\n\s*Step\s+\d+[:\.\-]|$))"
```
(1)

which achieves a verified $650/650$ (100%) step-count match against ground-truth `num_steps` across all MR-GSM8K splits (`verify_steps.py`). Given the fast tokenizer's character offset mapping $\{(c_t^{\text{start}}, c_t^{\text{end}})\}_{t=1}^{T}$ (`return_offsets_mapping=True`) and the character start index c_0 of Y_i within P_i, each step s_m with relative character span $[a_m, b_m)$ is mapped to the contiguous token index span $[t_m^{\text{start}}, t_m^{\text{end}})$ satisfying:

$$\mathcal{I}_m = \left\{ t \in \{1, \ldots, T\} \;\middle|\; c_t^{\text{start}} < c_0 + b_m \;\wedge\; c_t^{\text{end}} > c_0 + a_m \;\wedge\; c_t^{\text{start}} \neq c_t^{\text{end}} \right\}, \tag{2}$$

where $c_t^{\text{start}} \neq c_t^{\text{end}}$ filters out zero-width special control tokens.

Spatial Pooling Operators. Passing P_i through a frozen L-layer decoder-only Transformer (employing Grouped Query Attention, Rotary Position Embeddings [RoPE], and Pre-RMSNorm residual connections) yields per-layer hidden states $\mathbf{H}^{(\ell)} \in \mathbb{R}^{T \times d}$ for $\ell \in \{1, \ldots, L\}$. For each step $m \in \{1, \ldots, M_i\}$ with token indices $\mathcal{I}_m = [t_m^{\text{start}}, t_m^{\text{end}})$ and token length $|s_m| = t_m^{\text{end}} - t_m^{\text{start}}$, we compare three spatial compression operators:

$$\textbf{Mean Step Pooling:} \quad \mathbf{h}_{m,\text{mean}}^{(\ell)} = \frac{1}{|s_m|} \sum_{t=t_m^{\text{start}}}^{t_m^{\text{end}}-1} \mathbf{H}^{(\ell)}[t,:] \in \mathbb{R}^d, \tag{3}$$

$$\textbf{Last-Token Step Pooling:} \quad \mathbf{h}_{m,\text{last}}^{(\ell)} = \mathbf{H}^{(\ell)}[t_m^{\text{end}} - 1,:] \in \mathbb{R}^d, \tag{4}$$

$$\textbf{Global Whole-Response Baseline:} \quad \mathbf{h}_{m,\text{global}}^{(\ell)} = \frac{1}{M_i} \sum_{j=1}^{M_i} \mathbf{h}_{j,\text{mean}}^{(\ell)} \in \mathbb{R}^d. \tag{5}$$

Mean pooling tests whether evaluative error cues are distributed across semantic tokens within the step, Last-Token pooling tests whether causal self-attention concentrates step validity at the step-boundary delimiter token, and Global pooling replicates the whole-response representation of base INSPECTOR (Li et al., 2026) broadcast across step slots.

3.2 Probe Architectures and Multi-Layer Representation Dynamics

To test whether step-level deductive fallacies are linearly separable in the residual stream or require non-linear feature interaction, we implement three probe architectures mapping $\mathbf{h}_m^{(\ell)} \in \mathbb{R}^d$ to a scalar error logit $\hat{z}_m \in \mathbb{R}$:

1. Linear Probe (`LinearProbe`): A parameter-minimal single affine projection:

$$\hat{z}_m = \mathbf{w}^\top \mathbf{h}_m^{(\ell)} + b, \quad \mathbf{w} \in \mathbb{R}^d,\ b \in \mathbb{R}, \tag{6}$$

requiring only $d + 1$ trainable parameters (897 for `Qwen2.5-0.5B`; 1,025 for `Qwen3-0.6B`).

2. Non-Linear Bottleneck MLP Probe (`MLPProbe` / `StepProbe`): A 2-layer MLP incorporating bottleneck projection ($d \to d/4$), Layer Normalization, and GELU activation:

$$\hat{z}_m = \mathbf{w}_2^\top \,\mathrm{GELU}\Big(\mathrm{LayerNorm}\Big(\mathbf{W}_1 \mathbf{h}_m^{(\ell)} + \mathbf{b}_1\Big)\Big) + b_2, \tag{7}$$

where $\mathbf{W}_1 \in \mathbb{R}^{(d/4) \times d}$ and $\mathbf{w}_2 \in \mathbb{R}^{d/4}$, totaling $(d/4)(d+1) + 2(d/4) + (d/4+1)$ parameters (201,601 for $d = 896, d/4 = 224$; 263,169 for $d = 1024, d/4 = 256$).

3. Cross-Layer Attention Probe (`CLAP`): Rather than committing to a single layer ℓ, CLAP aggregates step representations $\{\mathbf{h}_m^{(\ell)}\}_{\ell=1}^{L}$ across all L layers using learnable layer-depth positional embeddings $\mathbf{e}_\ell \in \mathbb{R}^d$ and scaled dot-product attention with a learnable query $\mathbf{q} \in \mathbb{R}^d$:

$$\begin{aligned} \tilde{\mathbf{h}}_m^{(\ell)} &= \mathbf{h}_m^{(\ell)} + \mathbf{e}_\ell, \qquad \alpha_{m,\ell} = \frac{\exp\Big((\mathbf{W}_q \mathbf{q})^\top (\mathbf{W}_k \tilde{\mathbf{h}}_m^{(\ell)})/\sqrt{d}\Big)}{\sum_{j=1}^{L} \exp\Big((\mathbf{W}_q \mathbf{q})^\top (\mathbf{W}_k \tilde{\mathbf{h}}_m^{(j)})/\sqrt{d}\Big)}, \\ \mathbf{h}_m^{\text{fused}} &= \sum_{\ell=1}^{L} \alpha_{m,\ell} \mathbf{W}_v \tilde{\mathbf{h}}_m^{(\ell)}, \qquad \hat{z}_m = f_{\text{MLP}}\big(\mathbf{h}_m^{\text{fused}}\big). \end{aligned} \tag{8}$$

Because $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{d\times d}$ and $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{L\times d}$, CLAP contains 2,632,513 parameters on `Qwen2.5-0.5B` and 3,438,593 parameters on `Qwen3-0.6B` ($\sim 13\times$ larger than a single-layer `StepProbe`).

3.3 Causal Supervision Masking and Loss Formulation

In multi-step mathematical reasoning, steps form a directed causal chain. When a candidate solution makes its first error at step k^*, all subsequent steps $m \in \{k^*+1, \ldots, M_i\}$ are conditioned on a false intermediate premise. Consequently, post-error steps frequently contain locally valid arithmetic executed on corrupted numbers (e.g., correctly multiplying an erroneous subtotal from step k^*).

The Supervision Dilemma. If we train a step probe without causal masking, we must assign binary targets to post-error steps $m > k^*$:

- *Naive Unmasked (Target $= 0$):* Assigns $y_{k^*} = 1$ and $y_m = 0$ for all $m \neq k^*$. Because post-error steps $m > k^*$ carry residual distributional shifts and implausible quantities propagated from the error at k^*, forcing the probe to predict 0 on $m > k^*$ penalizes sensitivity to premise corruption and suppresses overall error recall.
- *Naive Unmasked (Target $= 1$):* Assigns $y_m = 1$ for all $m \geq k^*$ ($y_m = \max_{j\leq m} y_j$). This forces the probe to label locally sound arithmetic steps as errors, inflating the positive class prior and causing false-positive alarms on valid steps in correct solutions.

Causal Masked Step BCE Loss. To isolate the exact transition from valid deduction to logical failure, we define a causal binary mask $m_{i,m} \in \{0,1\}$ and target $y_{i,m} \in \{0,1\}$ for solution i with M_i valid steps and first error k_i^*:

$$m_{i,m} = \begin{cases} 1, & \text{if } k_i^* = 0 \text{ and } 1 \leq m \leq M_i, \\ 1, & \text{if } k_i^* > 0 \text{ and } 1 \leq m \leq k_i^*, \\ 0, & \text{if } (k_i^* > 0 \text{ and } m > k_i^*) \text{ or } m > M_i, \end{cases} \qquad y_{i,m} = \mathbb{I}[k_i^* > 0 \ \wedge \ m = k_i^*]. \tag{9}$$

We train the probe by minimizing the step-normalized **Causal Masked Step Binary Cross-Entropy Loss** (`MaskedStepBCELoss`, see Appendix C):

$$\mathcal{L}_{\text{causal}}(\theta) = \frac{\sum_{i=1}^{B}\sum_{m=1}^{M_{\max}} m_{i,m}\cdot[-y_{i,m}\log\sigma(\hat{z}_{i,m}) - (1-y_{i,m})\log(1-\sigma(\hat{z}_{i,m}))]}{\sum_{i=1}^{B}\sum_{m=1}^{M_{\max}} m_{i,m} + \epsilon}, \tag{10}$$

where $\sigma(\cdot)$ is the logistic sigmoid and $\epsilon = 10^{-4}$ prevents numerical underflow in FP16 tensors. Single-layer PyTorch probes are optimized using AdamW (lr $= 10^{-3}$, batch size 32, 20 epochs; lr $= 5\times10^{-4}$, 30 epochs for CLAP) with decoupled weight decay ($\lambda = 10^{-2}$ excluded from biases and LayerNorm weights) and gradient norm clipping at 1.0.

3.4 Evaluation Protocol, Metrics, and Diagnostic Audit Design

Inference Decoding Rule and Task Metrics. Given step error probabilities $\hat{p}_{i,m} = \sigma(\hat{z}_{i,m})$ for valid steps $1 \leq m \leq M_i$ and a decision threshold $\tau \in (0,1)$, the predicted first-error index is:

$$\hat{k}_i(\tau) = \begin{cases} \min\{m \in \{1,\ldots,M_i\} \mid \hat{p}_{i,m} > \tau\}, & \text{if } \exists m \leq M_i \text{ s.t. } \hat{p}_{i,m} > \tau, \\ 0, & \text{otherwise (predicted CORRECT).} \end{cases} \tag{11}$$

Following MR-GSM8K (Zeng et al., 2024), we evaluate three solution-level metrics over N test examples:

$$\text{Verdict Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\Big[(\hat{k}_i > 0) = (k_i^* > 0)\Big], \tag{12}$$

$$\text{First-Error Accuracy (FE-Acc)} = \frac{\sum_{i:k_i^*>0}\mathbb{I}\Big[\hat{k}_i = k_i^*\Big]}{\sum_{i=1}^{N}\mathbb{I}[k_i^* > 0]}, \tag{13}$$

$$\text{Joint Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\Big[\hat{k}_i = k_i^*\Big]. \tag{14}$$

Leakage-Free 5-Fold Out-of-Fold (OOF) Protocol. Because the validation split is small ($N_{\text{val}} = 50$, with only 25 faulty solutions, where a single solution shifts FE-Acc by 4.0 pp), selecting hyperparameters on validation alone exhibits high quantization variance, while selecting on the test split leaks information. For our primary diagnostic benchmark (Section 4.4), we pool Train + Val (550 solutions) and perform 5-fold stratified cross-validation (3 CV repeats = 15 fold predictors). All model fitting, layer selection ($\ell^* = \arg\max_\ell \text{AUROC}_{\text{OOF}}(\ell)$), and threshold selection (τ^*) use OOF predictions on the 550-example pool exclusively; the 100-example test split is scored once using the fold-ensemble.

Threshold-Free Metrics and Operating Point Degeneracy. On a class-balanced test split (50 correct, 50 faulty), a trivial classifier that sets $\tau \to 1$ (always predicting $\hat{k}_i = 0$, `CORRECT`) achieves 100% accuracy on correct solutions and 0% on faulty solutions, yielding **50.0% Joint Accuracy** with 0.0% First-Error Accuracy. Unless a model's exact first-error localizations outweigh its false rejections on correct solutions, optimizing τ solely for Joint Accuracy pushes τ upward to suppress flagging. We therefore prioritize two threshold-free metrics—**Step AUROC** on trusted steps ($m \leq k^*$, with and without trivial answer lines `####` / `The answer is`) and **Top-Step Accuracy** on known-faulty solutions ($\arg\max_{m \leq M_i} \hat{p}_{i,m} = k_i^*$)—and report thresholded metrics at two OOF-selected operating points: `balanced` ($\max_\tau \frac{1}{2}[\text{Verdict}(\tau) + \text{FE-Acc}(\tau)]$, introduced after preliminary runs that included test-split scores showed that the `joint` objective is degenerate for weak models on a 50/50 split, and then re-run from scratch under the OOF protocol) and `joint` ($\max_\tau \text{Joint}(\tau)$). All confidence intervals are 95% clustered bootstraps over solutions ($B = 2{,}000$ resamples).

3.5 Computational Complexity: Prefill Probing vs. Autoregressive Critique

Compared to generative critique tuning (where the model decodes up to $T_{\text{gen}} = 256$ new tokens greedily over a prompt of length T_{in}), Step-INSPECTOR eliminates the memory-bandwidth-bound autoregressive decoding phase entirely ($T_{\text{gen}} = 0$). Furthermore, because no tokens are generated, KV-cache allocation is disabled (`use_cache=False`), saving $2 \cdot L \cdot n_{\text{kv_heads}} \cdot d_{\text{head}} \cdot (T_{\text{in}} + T_{\text{gen}}) \cdot 2$ bytes of GPU memory per candidate. Once the single parallelizable prefill pass extracts step vectors up to layer ℓ^*, evaluating `StepProbe` requires $2\left(d \cdot \frac{d}{4} + \frac{d}{4}\right) \approx 0.40$M–0.52M FLOPs per step, or ≈ 6.8M–8.9M FLOPs per solution across $M_{\max} = 17$ step slots ($\sim 10^3\times$ fewer FLOPs than a single Transformer layer forward pass over the prompt), while guaranteeing 100% structural output validity without regex parsing failures.

4 Experiments and Results

Experimental Setup and Cross-Track Protocol Reconciliation. We evaluate on the course MR-GSM8K dataset (`assignment3_data.json`): 500 training, 50 validation, and 100 test solutions, each strictly balanced 50/50 between `CORRECT` and `INCORRECT` candidates ($M_{\max} = 17$ step slots in the cache). We extract FP16 representations from two frozen backbones: `Qwen/Qwen2.5-0.5B-Instruct` ($L = 24$, $d = 896$) and `Qwen/Qwen3-0.6B` ($L = 28$, $d = 1024$). Automated data audits (`m4_audit.py`) confirmed zero split leakage (`source_id`, `question`, and `candidate` overlaps all 0 across splits) and near-exact extraction parity (worst layer-wise cosine similarity ≥ 0.99999 between fresh forward passes and FP16 cached tensors across 20 re-extracted test solutions).

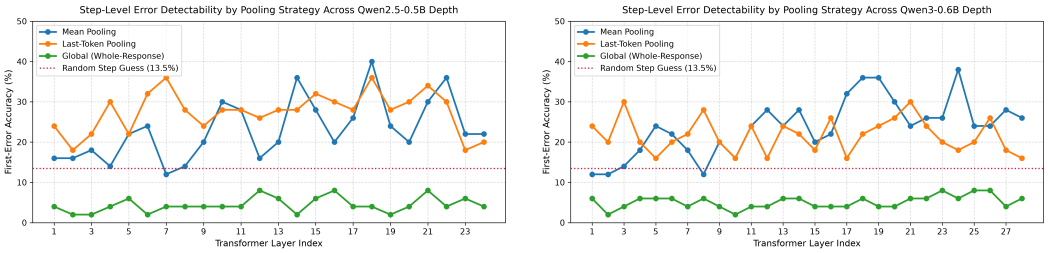
To isolate specific modeling factors, our four experimental tracks employ complementary training protocols that explain minor numerical differences at the same layer (e.g., Layer 19 on `Qwen2.5-0.5B`):

- **Track 1 (Section 4.1):** Uses class-balanced L_2-regularized Logistic Regression (`class_weight='balanced'`) with `StandardScaler` trained on `train` (500) and evaluated at $\tau = 0.5$. Because only 1 step per faulty solution has $y_{i,k^*} = 1$ while all preceding steps ($m < k^*$) and all steps of correct solutions have $y_{i,m} = 0$ (a $\sim 1:8$ positive-to-negative step ratio), `class_weight='balanced'` upweights positive steps by $\sim 8\times$. This shifts step probabilities above $\tau = 0.5$, yielding high raw First-Error Accuracy (up to 40.0%) but more false alarms on correct solutions (43.0%–50.0% Joint Accuracy).

- **Tracks 2 & 3 (Sections 4.2–4.3):** Train PyTorch probes (`LinearProbe`, `MLPProbe`, `CLAP`) on `train` (500) with early stopping on `val` (50) using unweighted $\mathcal{L}_{\text{causal}}$ and evaluate at uncalibrated $\tau = 0.5$. Because the step prior is unweighted, predicted probabilities at $\tau = 0.5$ are conservative (FPR 2.0%–12.0%), yielding high Joint Accuracy (59.0%–60.67%) alongside moderate uncalibrated FE-Acc (22.0% at L19; 36.0% at L22).
- **Track 4 (Section 4.4):** Trains a 15-model OOF ensemble (5 folds $\times$ 3 CV seeds) of `MLPProbe` on Train + Val (550) and calibrates τ^* via OOF optimization ($\tau_{\text{bal}} = 0.300$ on `Qwen2.5`, 0.225 on `Qwen3`), compensating for the unweighted step prior without sacrificing discrimination (34.0%–38.0% FE-Acc and 0.880/0.857 Step AUROC).

4.1 Track 1: Spatial Pooling Ablation Across Transformer Depth

We first evaluate how spatial token compression affects step error localization across all layers of both backbones (`spatial_pooling.py`, `plot_pooling_ablation.py`; complete layer-by-layer numbers in Appendix A).



(a) `Qwen2.5-0.5B-Instruct` (Layers 1–24) (b) `Qwen3-0.6B` (Layers 1–28)

Figure 1: First-Error Accuracy (%) across Transformer depth for Mean Pooling, Last-Token Pooling, and Global Whole-Response Pooling (base INSPECTOR baseline), compared against the expected random step guess baseline ($\mathbb{E}[1/M_i] = 13.5\%$).

Table 1: Spatial Pooling Ablation on the Test Split ($N = 100$, 50 faulty). We report performance at three reference depths: 5-Fold OOF-Selected Layer (L19 / L16), Single-Val Argmax Layer (L14 / L17), and Peak Test Layer (L18 / L24), plus the mean First-Error Accuracy across all L layers.

Backbone	Pooling Strategy	Layer (OOF / Val / Peak)	FE-Acc (%)	Verdict (%)	Joint (%)	All-Layer Mean FE (%)
`Qwen2.5-0.5B` ($L = 24, d = 896$)	Mean Pooling	19 / 14 / 18	24.0 / **36.0** / **40.0**	**71.0** / **66.0** / **67.0**	43.0 / **47.0** / **50.0**	23.1
	Last-Token	19 / 14 / 18	**28.0** / 28.0 / 36.0	67.0 / 63.0 / 66.0	38.0 / 45.0 / 42.0	**27.5**
	Global (Whole-Resp.)	19 / 14 / 12	2.0 / 2.0 / 8.0	69.0 / 60.0 / 66.0	**45.0** / 40.0 / 44.0	4.4
`Qwen3-0.6B` ($L = 28, d = 1024$)	Mean Pooling	16 / 17 / 24	22.0 / **32.0** / **38.0**	70.0 / **69.0** / **68.0**	**46.0** / **46.0** / **48.0**	**23.7**
	Last-Token	16 / 17 / 21	**26.0** / 16.0 / 30.0	**71.0** / **69.0** / 64.0	45.0 / 39.0 / 44.0	21.6
	Global (Whole-Resp.)	16 / 17 / 25	4.0 / 4.0 / 8.0	66.0 / 64.0 / 66.0	44.0 / 40.0 / 47.0	5.1

Findings (Figure 1 & Table 1):

1. **Mathematical and Empirical Collapse of Global Whole-Response Pooling:** While Global pooling achieves reasonable binary Verdict Accuracy (60.0%–73.0%), its First-Error Accuracy averages only **4.4**% on `Qwen2.5-0.5B` and **5.1**% on `Qwen3-0.6B` (never exceeding 8.0%), falling well below the 13.5% random step guess baseline. Because Global pooling assigns identical error probabilities ($\hat{p}_{i,1} = \cdots = \hat{p}_{i,M_i}$) to every step in solution i, the threshold decoding rule $\hat{k}_i = \min\{m \mid \hat{p}_{i,m} > \tau\}$ can *only ever output* $\hat{k}_i = 1$ when triggered. Since only 5 of the 50 faulty test solutions (10.0%) have their ground-truth first error at Step 1, Global pooling is mathematically upper-bounded at 10.0% First-Error Accuracy, showing that whole-response representations cannot localize multi-step reasoning failures.
2. **Depth-Dependent Crossover Between Last-Token and Mean Pooling:** In shallow layers (Layers 1–8), Last-Token pooling outperforms Mean pooling (e.g., Layer 7 on `Qwen2.5`: 36.0% vs. 12.0%; Layer 3 on `Qwen3`: 30.0% vs. 14.0%) because token representations in early layers are dominated by local lexical identity, whereas the step-final delimiter

token aggregates preceding step context via causal self-attention. In middle-to-late layers (Layers 14–28), once contextualized semantic features distribute across the step's tokens, **Mean Pooling achieves the highest peak localization** on both models ($\mathbf{40.0}\%$ at L18 on `Qwen2.5-0.5B` and $\mathbf{38.0}\%$ at L24 on `Qwen3-0.6B`).

4.2 Track 2: Probe Capacity Ablation and Confusion Diagnostics

To determine whether extracting mathematical error signals requires non-linear feature composition, we compare `LinearProbe` against the 2-layer `MLPProbe` across 3 random seeds ($\{42, 123, 2026\}$) at fixed late depth Layer 22 under $\mathcal{L}_{\text{causal}}$ at $\tau = 0.5$ (`run_role2_capacity.py`; per-seed breakdown in Appendix B). Table 2 reports the mean $\pm$ standard deviation across seeds alongside solution-level confusion-matrix diagnostics.

Table 2: Probe Capacity Ablation (`LinearProbe` vs. `MLPProbe`) at Layer 22 across 3 seeds ($\{42, 123, 2026\}$, mean $\pm$ SD). Positive class is `INCORRECT` solution (50 faulty, 50 correct). Exact Loc. denotes the number of faulty solutions (out of 50) where $\hat{k}_i = k_i^*$.

Backbone	Probe	Params	Verdict (%)	FE-Acc (%)	Joint (%)	Precision (%)	Recall (%)	FPR (%)	F1 (%)	Exact Loc. (/50)
`Qwen2.5-0.5B`	`LinearProbe`	897	67.33 ± 0.58	23.33 ± 3.06	$\mathbf{60.67 \pm 1.53}$	$\mathbf{94.82 \pm 0.15}$	36.67 ± 1.15	$\mathbf{2.00 \pm 0.00}$	52.88 ± 1.22	11.67 ± 1.53
	`MLPProbe`	201,601	$\mathbf{70.67 \pm 1.53}$	$\mathbf{32.00 \pm 5.29}$	60.00 ± 1.00	81.98 ± 3.12	$\mathbf{53.33 \pm 7.02}$	12.00 ± 4.00	$\mathbf{64.33 \pm 4.27}$	$\mathbf{16.00 \pm 2.65}$
	Paired Δ (MLP − Lin)	—	$+3.33 \pm 1.53$	$+\mathbf{8.67 \pm 7.02}$	-0.67 ± 1.53	-12.84 ± 3.13	$+16.67 \pm 7.02$	$+10.00 \pm 4.00$	$+11.45 \pm 4.26$	$+4.33 \pm 3.51$
`Qwen3-0.6B`	`LinearProbe`	1,025	69.33 ± 5.03	$\mathbf{24.67 \pm 7.57}$	58.33 ± 2.52	86.10 ± 5.82	46.67 ± 12.70	8.00 ± 5.29	59.61 ± 10.99	$\mathbf{12.33 \pm 3.79}$
	`MLPProbe`	263,169	$\mathbf{71.67 \pm 3.21}$	24.00 ± 6.00	$\mathbf{59.33 \pm 2.08}$	$\mathbf{90.33 \pm 2.59}$	$\mathbf{48.67 \pm 8.08}$	$\mathbf{5.33 \pm 2.31}$	$\mathbf{62.93 \pm 6.60}$	12.00 ± 3.00
	Paired Δ (MLP − Lin)	—	$+2.33 \pm 2.08$	-0.67 ± 4.62	$+1.00 \pm 1.00$	$+4.23 \pm 8.07$	$+2.00 \pm 6.00$	-2.67 ± 6.43	$+3.31 \pm 4.49$	-0.33 ± 2.31

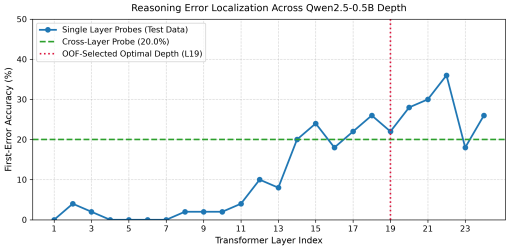
Findings (Table 2):

1. **Precision-Recall Trade-off on `Qwen2.5-0.5B`:** On `Qwen2.5-0.5B`, `LinearProbe` is highly conservative at $\tau = 0.5$, incurring only 1.00 false positive out of 50 correct solutions (FPR $= 2.00\%$, Precision $= 94.82\%$) and localizing the exact error step on $11.67/18.33 = 63.6\%$ of the faulty solutions it flags, but missing most errors (Recall $= 36.67\%$). Adding non-linear capacity (`MLPProbe`) increases error recall by $+16.67$ pp (to 53.33%), improving F1 by $+11.45$ pp and **First-Error Accuracy by** $+8.67$ **pp** (32.00% vs. 23.33%). However, because `MLPProbe` also raises FPR to 12.00%, Joint Accuracy remains virtually unchanged (60.00% vs. 60.67%).
2. **Linear Separability in `Qwen3-0.6B`:** On `Qwen3-0.6B`, the $1{,}025$-parameter `LinearProbe` already achieves 46.67% recall and 24.67% First-Error Accuracy, matching the $263{,}169$-parameter `MLPProbe` (24.00% FE-Acc; paired $\Delta = -0.67 \pm 4.62$ pp). This aligns with Kantamneni et al. (2025) and Li et al. (2026): increased probe capacity does not consistently improve overall performance when evaluative features are already linearly separable in the backbone's representations.

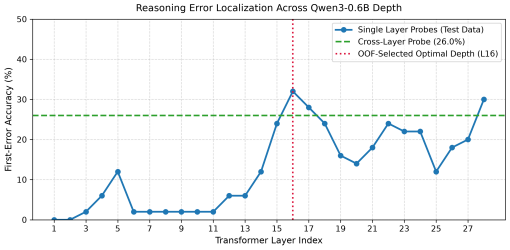
4.3 Track 3: Layer-Depth Geometry, Cross-Layer Attention, and Causal Masking

1. Layer-Depth Progression and Resolving Single-Split Spikes (Figure 2). Comparing the single-split thresholded curves ($\tau = 0.5$, Figures 2a–2b; full table in Appendix A) with the threshold-free 5-fold OOF Step AUROC sweep (Figure 2c) reveals why threshold-free OOF selection is essential:

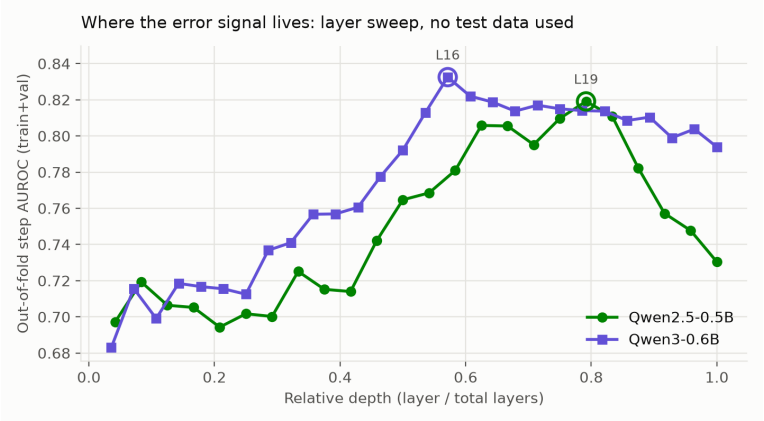
- *Early Layers (Relative Depth 0.0–0.4):* Under a fixed $\tau = 0.5$ threshold, Layers 1–11 produce near-0% First-Error Accuracy (50.0% Joint Accuracy) because early representations lack sharp confidence separation, keeping predicted probabilities below 0.5. Threshold-free OOF AUROC (Figure 2c) shows that discrimination in shallow layers is weak (AUROC $\approx$ 0.68–0.72, no better than surface length heuristics).
- *Mid-to-Late Plateau vs. Single-Split Spikes:* On the 50-example validation split (25 faulty solutions, 4.0 pp per solution), single-split validation FE-Acc peaks at Layer 14 (40.0%) on `Qwen2.5-0.5B` and Layer 17 (32.0%) on `Qwen3-0.6B`, while single-split test FE-Acc on `Qwen2.5-0.5B` shows an apparent spike at Layer 22 (36.0%). However, the 550-example 5-fold OOF Step AUROC curve (Figure 2c) demonstrates that this "Layer 22 spike" is simply small-sample threshold noise: in reality, evaluative signal plateaus broadly across middle-to-late layers ($\ell/L \in [0.57, 0.85]$). `Qwen3-0.6B` peaks earlier in relative depth at



(a) `Qwen2.5-0.5B` Test FE-Acc at $\tau = 0.5$ (`run_role3_depth.py`)



(b) `Qwen3-0.6B` Test FE-Acc at $\tau = 0.5$ (`run_role3_depth.py`)



(c) Threshold-Free 5-Fold Out-of-Fold (OOF) Step AUROC across Relative Depth (ℓ/L) on Train + Val ($N = 550$, zero test data used). Circles mark OOF-selected layers: L16 ($\ell/L = 0.57$) for `Qwen3-0.6B` and L19 ($\ell/L = 0.79$) for `Qwen2.5-0.5B`.

Figure 2: Layer-depth dynamics under thresholded single-split evaluation ($\tau = 0.5$, top; crimson dotted line marks the 5-fold OOF-selected layer L19 / L16, whereas the 50-example single-validation-split argmax layers are L14 / L17) versus threshold-free 5-fold OOF Step AUROC (bottom).

Layer 16 ($\ell/L = 0.57$, OOF AUROC $= 0.832$, Test FE-Acc $= 32.0\%$, Test Joint $= 60.0\%$) and stays flat to Layer 28, whereas `Qwen2.5-0.5B` peaks at **Layer 19** ($\ell/L = 0.79$, OOF AUROC $= 0.820$, Test FE-Acc $= 22.0\%$ at $\tau = 0.5$ and 38.0% at OOF $\tau = 0.45$) and falls off near the final output layers (Layers 23–24, AUROC $\rightarrow 0.730$).

2. Single-Layer Probes vs. Cross-Layer Attention Probe (CLAP). As shown in Figures 2a–2b, the parametric `CrossLayerAttentionProbe` achieves $\mathbf{20.0}\%$ Test FE-Acc (52.0% Joint Acc) on `Qwen2.5-0.5B` and $\mathbf{26.0}\%$ Test FE-Acc (57.0% Joint Acc) on `Qwen3-0.6B`, underperforming the OOF-selected single-layer probes (L19: 22.0% FE-Acc, 59.0% Joint on `Qwen2.5`; L16: 32.0% FE-Acc, 60.0% Joint on `Qwen3`). Because CLAP introduces 2.63M (`Qwen2.5`) to 3.44M (`Qwen3`) trainable parameters ($13\times$ more than `StepProbe`) trained on only 500 solutions, its cross-layer attention weights overfit and mix uninformative shallow-layer representations into the residual summary, whereas a single mid-to-late residual layer already accumulates the relevant deductive features via Pre-RMSNorm residual connections.

Table 3: Track 3 Ablation: Causal Loss Masking vs. Naive Unmasked Supervision ($\tau = 0.5$, seed 42). We report results at the 5-Fold OOF-Selected Layers (L19 for `Qwen2.5-0.5B`, L16 for `Qwen3-0.6B`) as well as at the `Qwen2.5-0.5B` single-split test peak (Layer 22).

Backbone & Layer	Supervision Masking Scheme	Verdict (%)	FE-Acc (%)	Joint (%)
`Qwen3-0.6B` (OOF Layer 16)	**Proposed Causal Masking ($\mathcal{L}_{\text{causal}}$)**	74.0	**32.0**	**60.0**
	Naive Unmasked (Target $= 0$)	65.0	12.0	54.0
	Naive Unmasked (Target $= 1$)	**77.0**	30.0	57.0
`Qwen2.5-0.5B` (OOF Layer 19)	**Proposed Causal Masking ($\mathcal{L}_{\text{causal}}$)**	65.0	22.0	**59.0**
	Naive Unmasked (Target $= 0$)	54.0	8.0	53.0
	Naive Unmasked (Target $= 1$)	**71.0**	**26.0**	56.0
`Qwen2.5-0.5B` (Layer 22)	**Proposed Causal Masking ($\mathcal{L}_{\text{causal}}$)**	**72.0**	**36.0**	**60.0**
	Naive Unmasked (Target $= 0$)	62.0	18.0	54.0
	Naive Unmasked (Target $= 1$)	71.0	30.0	57.0

3. Causal Supervision Masking Ablation (Table 3). Across both models and layer depths, **Proposed Causal Masking ($\mathcal{L}_{\text{causal}}$) consistently achieves the highest Joint Accuracy** (60.0% on `Qwen3` L16, 59.0% on `Qwen2.5` L19, and 60.0% on `Qwen2.5` L22):

- *Failure of Naive Unmasked (Target $= 0$):* Supervising post-error steps ($m > k^*$) as valid ($y_m = 0$) causes a severe collapse in First-Error Accuracy: dropping from $32.0\% \rightarrow 12.0\%$ (-20.0 pp) on `Qwen3-0.6B` L16, from $22.0\% \rightarrow 8.0\%$ (-14.0 pp) on `Qwen2.5-0.5B` L19, and from $36.0\% \rightarrow 18.0\%$ (-18.0 pp) on `Qwen2.5-0.5B` L22. Because post-error steps contain corrupted intermediate quantities, training the probe to output 0 on those steps suppresses its sensitivity to mathematical anomalies.
- *Trade-off of Naive Unmasked (Target $= 1$):* Labeling all post-error steps as faulty ($y_m = 1$ for $m \geq k^*$) inflates the positive label prior during training. At $\tau = 0.5$, this makes the probe over-trigger: while this aggressive flagging yields a higher raw Verdict/FE-Acc at `Qwen2.5` L19 ($71.0\%/26.0\%$ vs. $65.0\%/22.0\%$), it increases false alarms on valid steps and degrades Joint Accuracy across all three settings (57.0% vs. 60.0% on `Qwen3` L16; 56.0% vs. 59.0% on `Qwen2.5` L19; 57.0% vs. 60.0% on `Qwen2.5` L22). Causal masking cleanly resolves this dilemma by truncating supervision at k^*.

4.4 Track 4: Diagnostic Benchmarks, Shortcut Baselines, and Bias Audits

Finally, we subject Step-INSPECTOR to a rigorous meta-evaluation (`m4_evaluate.py`, `m4_stress_test.py`) comparing the 5-fold OOF ensemble probe against three content-free logistic regression baselines trained on identical causal masks: **Position Only** (one-hot step index m), **Length Only** ($\log(1 + |s_m|)$), and **Position + Length**. To preserve full readability, Table 4 reports the primary threshold-free metrics alongside the `balanced` operating point, Table 5 reports the `joint` operating point, and Table 6 reports the paired bootstrap differences.

1. What Works: Step-Level Discrimination Above Position and Length Shortcuts (Figure 3, Tables 4–6):

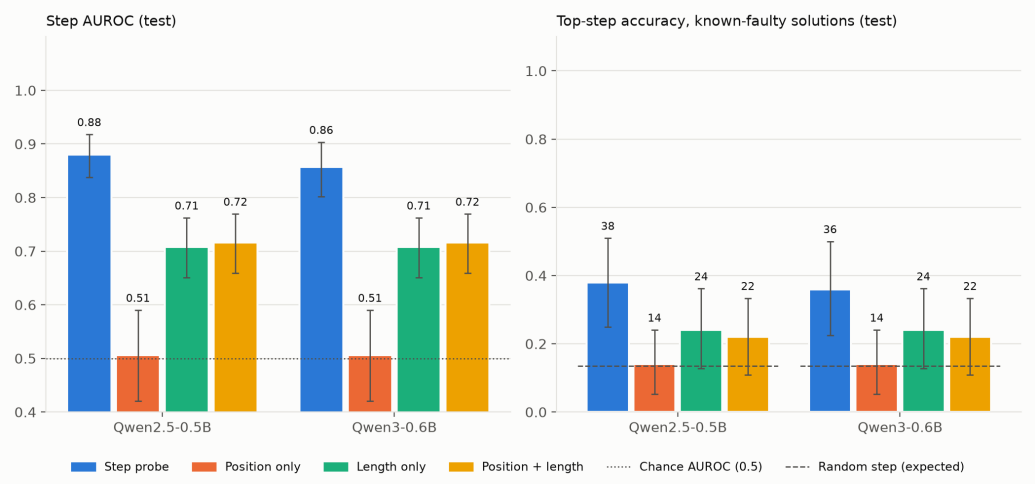


Figure 3: Primary threshold-free test results ($N = 100$ solutions, 50 faulty) with 95% solution-clustered bootstrap confidence intervals. **Left:** Step AUROC on trusted steps (note: y-axis starts at 0.40; dotted line marks 0.50 chance). **Right:** Top-Step Accuracy (%) on known-faulty solutions (y-axis starts at 0.0; dashed line marks 13.5% expected random step guess).

Table 4: Main Diagnostic Benchmark (Part 1: Threshold-Free Metrics and Balanced Operating Point) on the Test Split (100 solutions, 50 faulty; 95% bootstrap CIs in brackets). Layers (L19 for `Qwen2.5-0.5B`, L16 for `Qwen3-0.6B`) and thresholds τ_{bal} were selected via 5-fold OOF on Train + Val (550 solutions). Reference baselines: Expected Random Top-Step = 13.5%; Modal Step (Step 1) Top-Step = 10.0%; Always-Correct Joint = 50.0%.

		Threshold-Free (Primary)			Balanced Operating Point (τ_{bal})			
Model	**Method**	**Step AUROC**	**No-Ans AUROC**	**Top-Step (%)**	τ_{bal}	**Verdict (%)**	**FE-Acc (%)**	**Joint (%)**
`Qwen2.5-0.5B` (OOF L19)	**Step Probe**	**0.880** [0.84, 0.92]	**0.849**	**38.0** [25, 51]	0.300	**73.0** [64, 81]	**34.0** [21, 47]	**55.0** [45, 64]
	Position Only	0.506 [0.42, 0.59]	0.441	14.0 [5, 24]	0.075	50.0 [40, 60]	10.0 [2, 19]	5.0 [1, 10]
	Length Only	0.708 [0.65, 0.76]	0.626	24.0 [13, 36]	0.150	55.0 [45, 65]	16.0 [6, 27]	28.0 [20, 37]
	Pos. + Length	0.716 [0.66, 0.77]	0.636	22.0 [11, 33]	0.125	51.0 [41, 61]	18.0 [8, 29]	14.0 [7, 21]
`Qwen3-0.6B` (OOF L16)	**Step Probe**	**0.857** [0.80, 0.90]	**0.823**	**36.0** [22, 50]	0.225	**71.0** [62, 79]	**34.0** [20, 47]	**52.0** [42, 62]
	Position Only	0.506 [0.42, 0.59]	0.441	14.0 [5, 24]	0.075	50.0 [40, 60]	10.0 [2, 19]	5.0 [1, 10]
	Length Only	0.708 [0.65, 0.76]	0.626	24.0 [13, 36]	0.150	55.0 [45, 65]	16.0 [6, 27]	28.0 [20, 37]
	Pos. + Length	0.716 [0.66, 0.77]	0.636	22.0 [11, 33]	0.125	51.0 [41, 61]	18.0 [8, 29]	14.0 [7, 21]

Table 5: Main Diagnostic Benchmark (Part 2: Joint-Optimized Operating Point τ_{joint}, maximizing Joint Accuracy on OOF data).

Model	**Method**	τ_{joint}	**Verdict (%)**	**First-Error (%)**	**Joint (%)**
`Qwen2.5-0.5B` (OOF L19)	**Step Probe**	0.450	**74.0** [66, 82]	**38.0** [25, 52]	**61.0** [51, 70]
	Position Only	0.500	50.0 [40, 60]	0.0 [0, 0]	50.0 [40, 60]
	Length Only	0.550	50.0 [40, 60]	0.0 [0, 0]	50.0 [40, 60]
	Pos. + Length	0.650	50.0 [40, 60]	0.0 [0, 0]	50.0 [40, 60]
`Qwen3-0.6B` (OOF L16)	**Step Probe**	0.850	**68.0** [59, 77]	**14.0** [5, 24]	**53.0** [43, 63]
	Position Only	0.500	50.0 [40, 60]	0.0 [0, 0]	50.0 [40, 60]
	Length Only	0.550	50.0 [40, 60]	0.0 [0, 0]	50.0 [40, 60]
	Pos. + Length	0.650	50.0 [40, 60]	0.0 [0, 0]	50.0 [40, 60]

Table 6: Paired Bootstrap Differences: Step Probe − Baseline (95% CI clustered over solutions; an interval strictly excluding 0 indicates a statistically clear improvement).

Model	**Baseline**	**Δ Step AUROC**	**Δ Top-Step (pts)**	**Δ Bal. Verdict (pts)**	**Δ Bal. FE-Acc (pts)**
`Qwen2.5-0.5B`	Position Only	[**+0.286**, **+0.463**]	[**+8.0**, **+39.5**]	[**+10.0**, **+36.0**]	[**+7.4**, **+40.4**]
	Length Only	[**+0.109**, **+0.238**]	[−1.8, +28.9]	[**+5.0**, **+30.0**]	[**+4.7**, **+31.5**]
	Position + Length	[**+0.103**, **+0.227**]	[**+1.7**, **+30.0**]	[**+9.0**, **+35.0**]	[**+1.9**, **+31.4**]
`Qwen3-0.6B`	Position Only	[**+0.258**, **+0.440**]	[**+6.1**, **+38.0**]	[**+8.0**, **+34.0**]	[**+9.8**, **+38.8**]
	Length Only	[**+0.073**, **+0.222**]	[−2.1, +26.5]	[**+3.0**, **+29.0**]	[**+6.1**, **+30.4**]
	Position + Length	[**+0.065**, **+0.214**]	[+0.0, +27.5]	[**+7.0**, **+32.0**]	[**+2.6**, **+29.5**]

- *Step AUROC Significantly Beats All Content-Free Baselines:* Step-INSPECTOR achieves a Step AUROC of $\mathbf{0.880}$ $[0.84, 0.92]$ on `Qwen2.5-0.5B` and $\mathbf{0.857}$ $[0.80, 0.90]$ on `Qwen3-0.6B`. As shown in Table 6, paired bootstrap 95% CIs for ΔAUROC strictly exclude zero against **Position Only**, **Length Only**, and **Position + Length** on both backbones.
- *Signal Is Not Merely Final Answer-Line Detection:* Final answer lines (e.g., `Step M: #### 42`) are short and almost always valid, allowing Length-Only to reach 0.708 AUROC. Removing answer lines (`No-Ans AUROC`) drops Length-Only to 0.626, whereas the Step Probe maintains $\mathbf{0.849}$ (`Qwen2.5`) and $\mathbf{0.823}$ (`Qwen3`), showing genuine intermediate step discrimination.

2. What Not to Claim: Statistical Precision and Operating Point Caveats:

- *Top-Step Localization Beats Random and Position, But Its Gain Over Length Includes Zero:* On the 50 faulty test solutions, Top-Step Accuracy reaches 38.0% $[25, 51]$ (`Qwen2.5`) and 36.0% $[22, 50]$ (`Qwen3`), clearly exceeding the 13.5% random baseline and Position-Only (14.0%). However, because $N_{\text{faulty}} = 50$ yields $\sim \pm 13$ pp confidence intervals, the paired Top-Step difference against Length-Only (24.0%) spans $[-1.8, +28.9]$ pp on `Qwen2.5` and $[-2.1, +26.5]$ pp on `Qwen3` (Table 6). Because these intervals include zero, we cannot claim that the probe statistically beats step length at ranking the exact first-error step first, even though thresholded `balanced` FE-Acc (34.0% vs. 16.0%) has a positive paired CI.
- *Joint Accuracy at the Balanced Operating Point Does Not Beat the 50% Trivial Baseline:* At the `balanced` operating point ($\tau_{\text{bal}} = 0.300/0.225$), Joint Accuracy is 55.0% $[45, 64]$ (`Qwen2.5`) and 52.0% $[42, 62]$ (`Qwen3`), whose confidence intervals include the 50.0% trivial "always-correct" baseline. Under the `joint` operating point (Table 5), all three baselines degenerate to $\tau_{\text{joint}} \in [0.500, 0.650]$ (predicting `CORRECT` on 100% of solutions for 0.0% FE-Acc and 50.0% Joint); while `Qwen2.5-0.5B` reaches 61.0% $[51, 70]$ Joint Accuracy (38.0% FE-Acc), `Qwen3-0.6B` suffers from threshold instability ($\tau_{\text{joint}} = 0.850$, collapsing test FE-Acc to 14.0% for 53.0% $[43, 63]$ Joint).
- *No Winner Between Backbones:* The test AUROC difference between `Qwen2.5-0.5B` (0.880) and `Qwen3-0.6B` (0.857) is only 0.023, and their OOF AUROC ordering (0.832 `Qwen3` vs. 0.820 `Qwen2.5`) reverses on the test split.

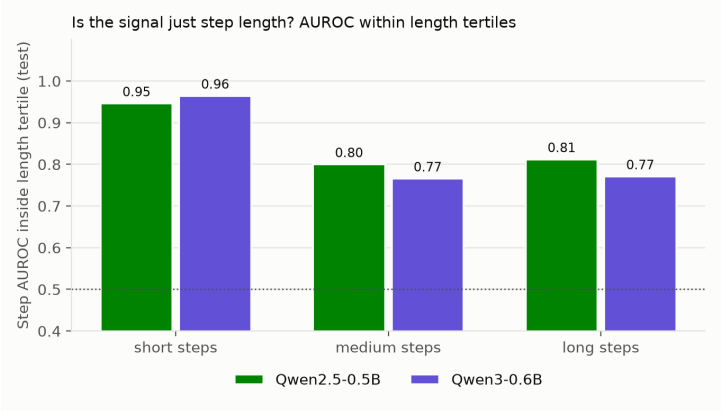


Figure 4: Observational Length Audit: Test Step AUROC stratified within step-length tertiles (short, medium, long non-answer trusted steps; note: y-axis starts at 0.40, dotted line marks 0.50 chance).

3. Observational Length Audit and Controlled Padding Stress Test (Figure 4 & Tables 7–8):

- *Within-Tertile Discrimination vs. Length Correlation:* On non-answer correct steps (Table 7), predicted error probability rises with step token length: Spearman $\rho = 0.36$ $[0.25, 0.46]$ on `Qwen2.5-0.5B` and $\rho = 0.37$ $[0.26, 0.49]$ on `Qwen3-0.6B`. However, within step-length tertiles (Figure 4), Step AUROC remains strong inside every third: $\mathbf{0.9463/0.8001/0.8122}$ (short/mid/long) on `Qwen2.5-0.5B` and

Table 7: Observational Length Audit of the Step Probe Across Step-Length Tertiles on Non-Answer Trusted Test Steps.

Model	ρ(**length**, $\hat{p}$) **Correct** [95% **CI**]	**AUROC Short**	**AUROC Mid**	**AUROC Long**
`Qwen2.5-0.5B` (OOF L19)	0.36 [0.25, 0.46]	0.9463	0.8001	0.8122
`Qwen3-0.6B` (OOF L16)	0.37 [0.26, 0.49]	0.9646	0.7659	0.7711

Table 8: Controlled Padding Stress Test (`m4_stress_test.py`) on $n = 20$ correct and $n = 20$ first-error arithmetic steps. We report mean added tokens, mean change in predicted error probability $\Delta\hat{p}$ [95% CI], and step-flag flips at the `balanced` threshold.

Model	Padding	Group ($n = 20$)	+Tok	Mean $\Delta\hat{p}$ [95% CI]	On→Off	Off→On
`Qwen2.5-0.5B`	`filler`	`correct`	+21	+0.149 [+0.091, +0.207]	0	5
	`filler`	`first_error`	+21	+0.169 [+0.098, +0.246]	0	6
	`plain`	`correct`	+15	+0.308 [+0.211, +0.412]	0	9
	`plain`	`first_error`	+15	+0.326 [+0.212, +0.430]	0	7
	`repeat`	`correct`	+32	+0.363 [+0.222, +0.505]	0	9
	`repeat`	`first_error`	+31	+0.312 [+0.172, +0.453]	1	8
`Qwen3-0.6B`	`filler`	`correct`	+21	+0.091 [+0.035, +0.159]	0	3
	`filler`	`first_error`	+21	+0.188 [+0.120, +0.255]	0	8
	`plain`	`correct`	+15	+0.083 [+0.035, +0.142]	0	3
	`plain`	`first_error`	+15	+0.165 [+0.106, +0.229]	0	4
	`repeat`	`correct`	+32	+0.003 [−0.028, +0.034]	0	1
	`repeat`	`first_error`	+31	+0.001 [−0.076, +0.073]	1	2

0.9646/0.7659/0.7711 on `Qwen3-0.6B`, indicating that the probe is not solely exploiting length.

- *Sensitivity to Textual Padding Interventions:* In Table 8, we append three length-increasing perturbations—`filler` (neutral explanation), `plain` (concise explanation), and `repeat` (repeating the step's own text)—to 20 correct and 20 first-error arithmetic steps. `Qwen2.5-0.5B` error scores rise systematically by $+0.149$ to $+0.363$ across all three paddings. In contrast, `Qwen3-0.6B` shifts by $+0.083$ to $+0.091$ on `filler`/`plain` (correct steps) and shows **no systematic shift for repeated step text** (mean $\Delta\hat{p} = +0.003$ $[-0.028, +0.034]$ on correct steps and $+0.001$ $[-0.076, +0.073]$ on first-error steps), although individual step scores moved in both directions (for example, on first-error steps the mean absolute change is 0.12, with $5/20$ steps falling by > 0.1 and $4/20$ rising by > 0.1). Because added text changes wording and semantics alongside token count, these results measure sensitivity to phrasing interventions rather than pure causal length bias.

5 Discussion and Limitations

1. **Sample Size and Statistical Resolution:** With 50 faulty test solutions, solution-clustered 95% bootstrap confidence intervals span roughly ± 13 percentage points. Differences smaller than this margin (such as the 2 pp Top-Step gap between backbones or minor AUROC variations across adjacent mid-to-late layers) lie within sampling noise.
2. **Disclosure of Operating Point Selection and Joint Degeneracy:** The `balanced` operating point was introduced after preliminary runs that included test-split scores showed that optimizing for `joint` accuracy is degenerate on a $50/50$ split whenever a model's exact first-error localizations do not outweigh its false alarms on correct solutions (as occurs for all three content-free baselines); the final evaluation was then re-run from scratch under the strict 5-fold OOF protocol, and we report both operating points transparently in Tables 4–5.
3. **Observational Length Correlations and Intervention Confounds:** Empirically, the probe assigns higher error scores to longer correct steps ($\rho = 0.36$–0.37); we hypothesize—though our experiments do not directly test—that this may reflect correlations between step length and multi-clause arithmetic complexity in the training distribution. Similarly, because our controlled padding interventions alter wording and discourse structure alongside token count,

they measure sensitivity to text perturbations rather than isolating length as a pure causal variable.

6 CONCLUSION

We presented **Step-INSPECTOR**, extending the *Representation-as-a-Judge* paradigm from coarse whole-response scoring to decoding-free, step-level mathematical error localization. Across `Qwen2.5-0.5B-Instruct` and `Qwen3-0.6B`, we showed that whole-response pooling washes out localized step errors, whereas token-aligned step-span mean pooling trained with **Causal Supervision Masking** ($\mathcal{L}_{\text{causal}}$) extracts strong step-level error signals from middle-to-late Transformer layers (Step AUROC $= 0.857$–0.880). Our diagnostic audits indicate that ultra-lightweight frozen SLMs encode step-level mathematical validity beyond simple position and length shortcuts in threshold-free step discrimination, while underscoring that establishing a statistically significant localization advantage over length heuristics requires larger faulty-solution sample sizes.

REPRODUCIBILITY AND AI USAGE STATEMENT

All code, deterministic regex step-slicing utilities, cached tensor extraction scripts, and multi-seed evaluation pipelines are provided in the accompanying submission archive and at `https://github.com/thejesshwar/step_inspector`. Automated audits (`m4_audit.py`) verify zero data leakage across splits and cosine similarity ≥ 0.99999 between fresh forward passes and FP16 cached tensors. In accordance with AI60213 course guidelines, generative AI tools were used strictly for conceptual brainstorming of the INSPECTOR extension, debugging assistance, and LaTeX formatting/polishing of the manuscript. All experimental designs, PyTorch modules, loss formulations, and statistical evaluations were implemented and verified by the four group members.

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A COMPLETE LAYER-BY-LAYER PROGRESSION TABLES

Tables 9 and 10 report the complete layer-by-layer First-Error Accuracy (`FE-Acc`), Verdict Accuracy (`V-Acc`), and Joint Accuracy (`J-Acc`) across all 24 layers of `Qwen2.5-0.5B-Instruct` and all 28 layers of `Qwen3-0.6B` from Track 1 (`spatial_pooling.py`) and Track 3 (`run_role3_depth.py`).

Table 9: Complete Layer-by-Layer Results on `Qwen2.5-0.5B-Instruct` (24 Layers, $d = 896$, $\tau = 0.5$). Bold indicates the 5-Fold OOF-Selected Layer (Layer 19).

	Mean Pool (Linear LR)			Last-Token (Linear LR)			Global Pool (Linear LR)			Mean Pool (StepProbe MLP)		
Layer	**V (%)**	**FE (%)**	**J (%)**	**V (%)**	**FE (%)**	**J (%)**	**V (%)**	**FE (%)**	**J (%)**	**Val FE (%)**	**Test FE (%)**	**Test J (%)**
01	60.0	16.0	35.0	58.0	24.0	23.0	60.0	4.0	43.0	0.0	0.0	50.0
02	57.0	16.0	31.0	56.0	18.0	23.0	53.0	2.0	36.0	4.0	4.0	52.0
03	66.0	18.0	35.0	59.0	22.0	27.0	57.0	2.0	41.0	0.0	2.0	51.0
04	65.0	14.0	34.0	56.0	30.0	31.0	64.0	4.0	43.0	0.0	0.0	50.0
05	65.0	22.0	39.0	59.0	22.0	32.0	62.0	6.0	43.0	0.0	0.0	50.0
06	63.0	24.0	41.0	61.0	32.0	39.0	62.0	2.0	43.0	0.0	0.0	50.0
07	63.0	12.0	36.0	61.0	36.0	39.0	60.0	4.0	41.0	0.0	0.0	50.0
08	64.0	14.0	36.0	58.0	28.0	35.0	64.0	4.0	42.0	0.0	2.0	51.0
09	65.0	20.0	37.0	63.0	24.0	35.0	61.0	4.0	43.0	0.0	2.0	51.0
10	68.0	30.0	44.0	57.0	28.0	33.0	73.0	4.0	45.0	0.0	2.0	51.0
11	68.0	28.0	42.0	69.0	28.0	44.0	72.0	4.0	46.0	0.0	4.0	52.0
12	69.0	16.0	40.0	69.0	26.0	41.0	66.0	8.0	44.0	0.0	10.0	55.0
13	67.0	20.0	42.0	66.0	28.0	43.0	66.0	6.0	46.0	0.0	8.0	54.0
14	66.0	36.0	47.0	63.0	28.0	45.0	60.0	2.0	40.0	40.0	20.0	55.0
15	72.0	28.0	47.0	58.0	32.0	37.0	65.0	6.0	44.0	16.0	24.0	57.0
16	70.0	20.0	41.0	65.0	30.0	40.0	64.0	8.0	43.0	8.0	18.0	56.0
17	73.0	26.0	46.0	69.0	28.0	40.0	59.0	4.0	39.0	16.0	22.0	59.0
18	67.0	40.0	50.0	66.0	36.0	42.0	62.0	4.0	43.0	16.0	26.0	60.0
19	**71.0**	**24.0**	**43.0**	**67.0**	**28.0**	**38.0**	**69.0**	**2.0**	**45.0**	**12.0**	**22.0**	**59.0**
20	72.0	20.0	42.0	68.0	30.0	42.0	65.0	4.0	44.0	24.0	28.0	60.0
21	68.0	30.0	47.0	63.0	34.0	40.0	67.0	8.0	44.0	28.0	30.0	60.0
22	69.0	36.0	50.0	70.0	30.0	42.0	62.0	4.0	42.0	28.0	36.0	60.0
23	65.0	22.0	41.0	64.0	18.0	33.0	62.0	6.0	42.0	12.0	18.0	54.0
24	69.0	22.0	42.0	66.0	20.0	36.0	54.0	4.0	36.0	20.0	26.0	55.0

Table 10: Complete Layer-by-Layer Results on `Qwen3-0.6B` (28 Layers, $d = 1024$, $\tau = 0.5$). Bold indicates the 5-Fold OOF-Selected Layer (Layer 16).

	Mean Pool (Linear LR)			Last-Token (Linear LR)			Global Pool (Linear LR)			Mean Pool (StepProbe MLP)		
Layer	**V (%)**	**FE (%)**	**J (%)**	**V (%)**	**FE (%)**	**J (%)**	**V (%)**	**FE (%)**	**J (%)**	**Val FE (%)**	**Test FE (%)**	**Test J (%)**
01	55.0	12.0	27.0	60.0	24.0	30.0	57.0	6.0	40.0	0.0	0.0	50.0
02	60.0	12.0	29.0	64.0	20.0	32.0	55.0	2.0	40.0	0.0	0.0	50.0
03	67.0	14.0	36.0	61.0	30.0	36.0	58.0	4.0	42.0	4.0	2.0	50.0
04	65.0	18.0	38.0	61.0	20.0	34.0	62.0	6.0	42.0	8.0	6.0	52.0
05	64.0	24.0	37.0	58.0	16.0	33.0	63.0	6.0	44.0	24.0	12.0	51.0
06	58.0	22.0	35.0	54.0	20.0	29.0	62.0	6.0	41.0	0.0	2.0	51.0
07	64.0	18.0	37.0	54.0	22.0	32.0	64.0	4.0	45.0	0.0	2.0	51.0
08	72.0	12.0	40.0	60.0	28.0	41.0	70.0	6.0	47.0	0.0	2.0	51.0
09	64.0	20.0	38.0	64.0	20.0	40.0	61.0	4.0	43.0	0.0	2.0	51.0
10	64.0	16.0	41.0	63.0	16.0	39.0	61.0	2.0	42.0	0.0	2.0	51.0
11	67.0	24.0	42.0	63.0	24.0	39.0	59.0	4.0	38.0	0.0	2.0	51.0
12	65.0	28.0	44.0	63.0	16.0	38.0	63.0	4.0	41.0	0.0	6.0	53.0
13	71.0	24.0	45.0	75.0	24.0	44.0	76.0	6.0	52.0	0.0	6.0	53.0
14	64.0	28.0	43.0	68.0	22.0	39.0	68.0	6.0	45.0	4.0	12.0	55.0
15	73.0	20.0	47.0	73.0	18.0	46.0	63.0	4.0	41.0	24.0	24.0	56.0
16	**70.0**	**22.0**	**46.0**	**71.0**	**26.0**	**45.0**	**66.0**	**4.0**	**44.0**	**24.0**	**32.0**	**60.0**
17	69.0	32.0	46.0	69.0	16.0	39.0	64.0	4.0	40.0	32.0	28.0	58.0
18	69.0	36.0	46.0	67.0	22.0	43.0	68.0	6.0	44.0	24.0	24.0	57.0
19	70.0	36.0	49.0	65.0	24.0	42.0	63.0	4.0	43.0	12.0	16.0	55.0
20	72.0	30.0	47.0	62.0	26.0	40.0	66.0	4.0	44.0	8.0	14.0	55.0
21	64.0	24.0	37.0	64.0	30.0	44.0	63.0	6.0	42.0	12.0	18.0	57.0
22	73.0	26.0	45.0	62.0	24.0	39.0	66.0	6.0	43.0	20.0	24.0	60.0
23	70.0	26.0	41.0	66.0	20.0	39.0	67.0	8.0	45.0	20.0	22.0	59.0
24	68.0	38.0	48.0	66.0	18.0	41.0	66.0	6.0	45.0	20.0	22.0	58.0
25	65.0	24.0	41.0	63.0	20.0	39.0	66.0	8.0	47.0	0.0	12.0	55.0
26	68.0	24.0	43.0	71.0	26.0	46.0	64.0	8.0	45.0	12.0	18.0	54.0
27	68.0	28.0	44.0	72.0	18.0	45.0	65.0	4.0	43.0	20.0	20.0	56.0
28	69.0	26.0	45.0	63.0	16.0	40.0	66.0	6.0	45.0	20.0	30.0	60.0

B Per-Seed Confusion-Matrix Diagnostics (Track 2)

Table 11 reports all 12 individual training runs (3 seeds $\times$ 2 probe architectures $\times$ 2 backbones) from Track 2 (`run_role2_capacity.py`) at Layer 22 ($\tau = 0.5$).

Table 11: Per-Seed Probe Capacity Ablation and Confusion-Matrix Diagnostics at Layer 22 ($N = 100$ test solutions: 50 faulty, 50 correct).

Backbone	Probe	Seed	Params	TP	TN	FP	FN	Prec. (%)	Rec. (%)	FPR (%)	F1 (%)	Verdict (%)	FE-Acc (%)	Joint (%)
Qwen2.5-0.5B	LinearProbe	42	897	18	49	1	32	94.74	36.0	2.0	52.17	67.0	20.0 (10/50)	59.0
	LinearProbe	123	897	19	49	1	31	95.00	38.0	2.0	54.29	68.0	26.0 (13/50)	62.0
	LinearProbe	2026	897	18	49	1	32	94.74	36.0	2.0	52.17	67.0	24.0 (12/50)	61.0
	MLPProbe	42	201,601	30	42	8	20	78.95	60.0	16.0	68.18	72.0	36.0 (18/50)	60.0
	MLPProbe	123	201,601	27	44	6	23	81.82	54.0	12.0	65.06	71.0	34.0 (17/50)	61.0
	MLPProbe	2026	201,601	23	46	4	27	85.19	46.0	8.0	59.74	69.0	26.0 (13/50)	59.0
Qwen3-0.6B	LinearProbe	42	1,025	27	43	7	23	79.41	54.0	14.0	64.29	70.0	30.0 (15/50)	58.0
	LinearProbe	123	1,025	16	48	2	34	88.89	32.0	4.0	47.06	64.0	16.0 (8/50)	56.0
	LinearProbe	2026	1,025	27	47	3	23	90.00	54.0	6.0	67.50	74.0	28.0 (14/50)	61.0
	MLPProbe	42	263,169	25	48	2	25	92.59	50.0	4.0	64.94	73.0	24.0 (12/50)	60.0
	MLPProbe	123	263,169	20	48	2	30	90.91	40.0	4.0	55.56	68.0	18.0 (9/50)	57.0
	MLPProbe	2026	263,169	28	46	4	22	87.50	56.0	8.0	68.29	74.0	30.0 (15/50)	61.0

C Causal Masked Loss Implementation and Probe Specifications

Listing 1 shows the PyTorch implementation of `MaskedStepBCELoss` (`custom_loss.py`), which truncates gradient flow after the first erroneous step k^* while maintaining FP16 stability via $\epsilon = 10^{-4}$. Table 12 summarizes the exact parameter counts across all probe architectures.

Listing 1: PyTorch implementation of Causal Masked Step BCE Loss (`custom_loss.py`).

```
class MaskedStepBCELoss(nn.Module):
    """
    Causal Masked Step Binary Cross-Entropy Loss:
    L = [ sum_{m=1}^M m_m * BCE(z_m, y_m) ] / [ sum_{m=1}^M m_m + eps ]
    where m_m = 1.0 for valid deductive steps up to the first error (m <= k*)
    and m_m = 0.0 for post-error steps (m > k*) and padded slots.
    """
    def __init__(self, eps: float = 1e-4):
        super().__init__()
        self.eps = eps

    def forward(self, logits: torch.Tensor, targets: torch.Tensor, masks: torch.Tensor) ->
         torch.Tensor:
        targets = targets.view_as(logits).to(dtype=logits.dtype)
        masks = masks.view_as(logits).to(dtype=logits.dtype)
        bce = F.binary_cross_entropy_with_logits(logits, targets, reduction='none')
        return (bce * masks).sum() / (masks.sum() + self.eps)
```

Table 12: Trainable Parameter Count Breakdown Across Probe Architectures ($L = 24, d = 896$ for `Qwen2.5-0.5B`; $L = 28, d = 1024$ for `Qwen3-0.6B`).

Architecture	Analytical Formula	`Qwen2.5-0.5B`	`Qwen3-0.6B`
`LinearProbe`	$d + 1$	897	1,025
`MLPProbe` (`StepProbe`)	$\frac{d}{4}(d+1) + \frac{d}{2} + \frac{d}{4} + 1$	201,601	263,169
`CrossLayerAttentionProbe`	$Ld + d + 3d^2 +$ MLPProbe	2,632,513	3,438,593